TECHMEN 5669 · Driver & Operator Controls

Robot code: main, Oct 27 2026 · Driver = Port 0 · Operator = Port 1



Driver · Port 0 · driving, intake, climb

Left Stick	STICK	Drive (forward, back, left, right)
Right Stick	STICK	Turn the robot
RB or Start	HOLD	Slow mode (RB keeps your thumb on the turn stick)
LB	HOLD ½ S	Reset gyro (practice only). **Robot straight away from your DS first**. Buzz = done
RT	HOLD	Intake roller: balls IN
LT	HOLD	Intake roller: balls OUT (spit them back)
A	PRESS	Intake down / up
D-pad Up / Down	HOLD	Climber up / down (stops by itself at the ends)
D-pad Left	PRESS	Shooter idle at 2000 RPM on / off
Y	PRESS	**Defense mode** on / off: intake up, hood down, quicker drive, wheels lock in an X when you let go
Left Stick click	HOLD	Full drive power in vacuum / dump mode
X	·	Nothing

Operator · Port 1 · aiming and shooting

RB	PRESS	**Auto-aim on / off**. It turns ON by itself at teleop start (1 buzz = on, 2 = off)
RT	HOLD	Shoot. Fires when ready (buzzes). Aim OFF: spins up, then feeds
LT	HOLD	Spin up the shooter (aim ON: RT does it itself)
Y	PRESS	Hood up / down
D-pad Up / Down	HOLD	Move the hood up / down slowly
D-pad Left / Right	HOLD	Turn the turret left / right
Left Stick up / down	HOLD	Move the intake up / down slowly (backup)
B	HOLD	Spindex backwards (unjam)
A / X / LB	HOLD ½ S	Backup shots: A = Midshot, X = Left trench, LB = Right trench
Back	PRESS	**Vacuum mode** on / off: intake + auto-aim, fires by itself (scores in our zone, passes from the neutral zone)
Left Stick click	PRESS	**Dump mode** on / off: roller OUT (X-take) + auto-pass. Intake down first
Right Stick click	PRESS	Trench shot (hood up, robot stays put)
Start	PRESS	Target: AUTO → SCORE → PASS

Turret jog, hood jog or Y turns auto-aim OFF. Press RB to turn it back on.

TECHMEN 5669 · How To Use It

Keep this with the driver station

Before every match or practice

1. **Robot OFF.** Turret on the **HOME mark**. Hood **pushed all the way down** by hand. Intake **pushed all the way up**. Climber **all the way down**.
2. **Turn the robot on** while holding the hood down.
3. Driver Station **USB tab**: driver controller in **slot 0**, operator in **slot 1**. Press a button to see which row lights up; drag to fix. `F1` = rescan.
4. Driver Station **Team Station**: pick **our alliance color** (Red / Blue). At competition the field sets it.
5. Elastic: **"Pit Check"** should say **ALL GO**.
6. Practice only (a match sets it): robot **straight away from your driver station**, hold **Driver LB** until it buzzes.
7. Shop field: **"Practice Field"** switch ON. **Competition: OFF.**

Emergency keys (Driver Station laptop)

- `Enter` = **Disable** (stops the robot, no restart needed). Use this one.
- `Space` = **Emergency stop**. The robot must be powered off and on again.

What the buzzes mean

- **1 short** = turned ON (auto-aim, gyro reset, backup spot). **2 short** = turned OFF.
- **1 long** (both) = vacuum or dump started: the intake drops.
- Operator **steady** = ready to fire. **3 long** = hopper looks empty: go get balls.
- Operator **1 / 2 / 3 short** after Start = AUTO / SCORE / PASS.
- Driver **low hum** = our hub is open. **Pulses** = hub swap in 5 s.
- Driver **fast pulses** = pin (back off). **Steady** in the last 30 s = their tower, don't touch.
- **Both pulse** early in teleop = no FMS game data: coach sets "Auto Winner".

Shooting with auto-aim (normal way)

1. Auto-aim turns ON by itself when teleop starts (the turret follows our hub; the hood stays down until you shoot).
2. Operator **holds RT**. The shooter spins up to the right speed for the distance and fires each ball the moment the shot is ready. The controller **buzzes** while it's ready.
3. While scoring, the robot slows to shooting speed and locks its wheels when you let go of the sticks. Driver left stick click = full speed. If the hub is behind the turret, the robot turns itself a little (the turn stick always wins).

Shooting without auto-aim

1. Aim with **D-pad Left / Right** (turret) and **D-pad Up / Down** or **Y** (hood).
2. Hold **RT**: the shooter spins up and RT feeds once it is at speed (LT pre-spins).

Backup shots (if the camera is having trouble)

1. Park on the spot: **Midshot** (between our hub and our climb), **Left trench** or **Right trench** (left / right as seen from our driver station).
2. Operator **holds A** (Midshot), **X** (Left trench) or **LB** (Right trench) for half a second (buzz = set).
3. Hold **RT**.

Live version on any driver station laptop: **tools/controller-map/Start Controller Map.bat** (lights up each button as you press it). Full details: CLAUDE.md, "Controller Bindings".